AHWR Digital Twin

Production rollout execution plan, ownership model, and rig-edge / central hardware requirements

Purpose: A practical handoff document for the teams responsible for OT/PLC access, rig hardware, networking and cybersecurity, edge deployment, central infrastructure, application configuration, testing, training, and ongoing support.

Document field	Value
Scope	Production rollout to 50 rigs with one standardized indoor edge system per rig and an HA central CRMF
Deployment principle	Monitoring only; no command or control write-back to PLCs
Planning horizon	Wave-based production rollout; rig operating schedules and approved maintenance windows control duration
Prepared	27 July 2026
Status	Production rollout baseline — proving completed; apply per-rig commissioning and wave acceptance controls

This plan is based on the repository's proposal, deployment guide, Docker/Helm/Ansible assets, and current application status. Site drawings, ONGC standards, approved vendors, cybersecurity policy, and actual tag volumes remain controlling inputs.

1. Executive implementation decision

Proceed directly with production rollout in two controlled layers. Deploy the production HA central CRMF for 50 rigs, then install the standardized indoor edge package in approved waves. Each rig remains locally autonomous during WAN loss and uses authenticated outbound store-and-forward. The proven design is fixed; site-specific tag mapping, installation, commissioning, and acceptance remain mandatory for every rig.

Non-negotiable safety boundary: The solution receives and displays telemetry only. Do not create any PLC write route, control command, remote actuation, or central-to-rig operational path. ESD, lockout, BOP, accumulator, and related states are read-only alarms.

2. Delivery outcomes and scope boundaries

- Rig-edge HMI with live, correctly scaled telemetry, local historian, alarms, workover activity, reports, maintenance, user roles, and honest stale/no-data indication.
- Offline resilience: local HMI remains available during WAN loss; queued central batches replay oldest-first after restoration.
- Central CRMF with rig registry, authenticated ingestion, fleet overview, per-rig drill-down, data-quality monitoring, cross-rig alarms, reports, governance, audit, and operations monitoring.
- Repeatable installation pack: approved hardware BoQ, panel/network drawings, golden software image, tag workbook, test evidence, training records, rollback procedure, and support contacts.

Excluded from the production release:

- ETP/WITSML central server, ML inference, Kafka derived-channel processor, and standalone MQTT-to-Kafka bridge where repository manifests are explicitly scaffolded or set to zero replicas.
- Historical scrub-back replay UI, full ETP Avro encoding, direct SMS provider, and any unimplemented safety logic such as kick detection.
- Any claim that missing BOP/ESD or other critical field tags are validated merely because synthetic demo data is visible.

3. Hardware requirements

3.1 Rig-edge hardware — one workover rig

Item	Minimum / mandatory specification	Recommended production choice	Qty.
Industrial edge computer	Indoor industrial or enterprise-grade edge PC; 4 CPU cores; 16 GB RAM; 512 GB enterprise/industrial SSD; dual Ethernet; TPM 2.0; approved AC or 24 V DC supply	4–8 cores, 16–32 GB RAM, 1 TB high-endurance SSD; vendor lifecycle ≥5 years	1
Operator HMI	23.8–24 inch Full HD monitor for indoor control-room use	Commercial/industrial touch display if operator workflow requires touch; second display optional	1
Industrial switch	8-port managed Ethernet switch; VLAN, RSTP and port security	SNMP/syslog, spare ports and approved redundant power where available	1
Firewall / VPN router	Approved firewall/router; VPN, NAT and ACL; separate OT and WAN interfaces	HA-capable model where the rig network design requires device redundancy	1
Protocol gateway	OPC UA / Modbus / S7 gateway when direct read-only PLC access is not approved	Dual-port isolated gateway with store diagnostics	As needed
Serial converter	Isolated RS-485/Ethernet converter for legacy Modbus RTU	Industrial temperature range and surge isolation	As needed
UPS	Online UPS; ≥30 minutes for edge PC, display, switch and router	Monitored UPS with replaceable battery and graceful-shutdown contact	1
Panel enclosure	Lockable ventilated indoor cabinet/rack; MCB, surge protection, earthing and cable management	Panel temperature and power-health monitoring; forced ventilation only if heat study requires it	1
Keyboard / pointing device	Standard business-grade keyboard and pointing device	Sealed type only where housekeeping or spill risk requires it	1
Time source	ONGC NTP	GPS/NTP appliance only where network time is unavailable	Optional
Cold standby	Required rollout resilience item	Pre-imaged spare edge PC or recoverable golden SSD	1 per rollout group

Sizing basis: Indoor installation; fewer than 500 tags per rig at 1-second sampling; 90 days local raw-data retention. This ceiling equals fewer than 43.2 million values/day and fewer than 3.89 billion values over 90 days per rig. Use a 1 TB high-endurance SSD as the rollout standard unless the accepted proving data supports a smaller device.

3.2 Central production workload — 50 rigs

Component	Baseline	Notes
Workload ceiling	50 rigs × <500 tags × 1 Hz = <25,000 values/second	<2.16 billion values/day when every edge value is centralized
Recommended retention tier	90 days central raw 1 Hz + 5 years of 1-minute aggregates	Keeps operational replay while controlling long-term volume
Availability target	RKE2 HA across three failure domains; central availability ≥99.5%	No single control-plane, worker, database or ingress failure stops service
Network	Redundant data-centre LAN; TLS ingress; VPN/private WAN; 1/10 GbE server links	Publish portal and ingest through controlled gateway; never expose the database

3.3 Central production HA hardware baseline

Sizing basis: Fixed production ceiling of 50 rigs and fewer than 25,000 values/second. Storage assumes 90 days of raw 1 Hz telemetry at a conservative 8–20 compressed bytes/value, three database copies, WAL and operating headroom. Substitute the accepted measured compression figure if the completed proving record is more conservative.

Tier	Recommended node specification	Qty.	Purpose / storage
RKE2 control plane	8 vCPU, 32 GB RAM, 200 GB enterprise SSD, dual 10 GbE	3	etcd/control plane; spread across failure domains
General worker	16 vCPU, 64 GB RAM, 500 GB OS/container SSD, dual 10 GbE	3	App, ingress, identity, MQTT and monitoring; ≥30% headroom
Data worker	24 vCPU, 128 GB RAM, 1 TB local NVMe plus CSI-backed enterprise storage	3	Dedicated TimescaleDB/Kafka placement and storage bandwidth
TimescaleDB persistent storage	5 TB data + 500 GB WAL per database instance	3 sets	≈16.5 TB allocated primary-site storage including three copies
Kafka storage (if enabled)	500 GB enterprise SSD per broker	3	Replication factor 3; min in-sync replicas 2
Backup / object storage	20 TB usable protected object storage; expandable	1 service	Database backups, WAL archive, reports and recovery headroom
DR site	Compute and storage capable of the same 50-rig database workload	1 site	At least two DB replicas; size to approved RPO/RTO and retention
Load balancer / VIP	Redundant L4/L7 load balancer or approved managed equivalent	2 or managed	RKE2 API and application ingress

Production platform minimum service shape:

- CRMF backend: start with 3 replicas and autoscale to 12; frontend: 3 replicas. Preserve at least 30% cluster headroom.

- TimescaleDB: 3 instances; production override target per instance: 8 CPU / 32 GiB requested and up to 16 CPU / 64 GiB limited.
- Kafka, when enabled: 3 brokers; target at least 2 CPU / 8 GiB and 500 GB enterprise storage per broker.
- Keycloak: 2 instances plus a 2-instance PostgreSQL cluster; Timescale connection pooler: 2 instances.
- EMQX, when MQTT is enabled: use three instances with discovery; do not use the single-instance repository default in production.

3.4 Fixed production sizing inputs and controls

Input	Owner	Required decision
Active rigs	Program PMO	50 production rigs
Tags and edge retention	OT + data engineering	<500 tags/rig, 1 Hz, 90-day edge raw retention
Central ingest ceiling	Central platform	<25,000 values/s; <2.16 billion values/day
Central retention	Data owner / DGC	90-day raw 1 Hz + 5-year 1-minute aggregate baseline
Report/object volume	Application owner	20 TB protected object/backup tier with expansion path
Backup RPO/RTO and DR	Business continuity	Apply approved production RPO/RTO and recovery-test schedule
Portal performance	Operations	50-rig dashboard SLA: normal filters load in <5 seconds

4. Step-by-step implementation plan

Completed inputs: The production tag set has been identified and the edge-system configuration baseline has been finalized. The rollout must deploy that approved baseline; tag or configuration changes are handled only as controlled rig-specific exceptions.

Step 0 — Release the production rollout baseline

Freeze the 50-rig wave schedule, approved hardware/software release, site owners, maintenance windows and change-control route. **Exit:** Wave authorization, rig list, accountable owners and release version are signed.

Step 1 — Confirm site readiness

Confirm the indoor cabinet/rack, power, UPS, earthing, LAN/WAN, fixed IPs, DNS/NTP, PLC route and installation access for each rig. **Exit:** The site-readiness checklist and installation window are approved.

Step 2 — Bring the production CMF foundation online

Deploy the Section 3 HA compute, storage, load balancer, RKE2, TLS, identity, backup/DR and monitoring baseline. **Exit:** Cluster health, failover, backup/restore and security controls pass.

Step 3 — Stage and test edge kits

Load the golden OS and signed application release, apply host hardening, verify UPS/runtime health and record each asset/version. **Exit:** Each kit passes FAT and the spare/recovery image is usable.

Step 4 — Pre-register rigs and credentials

Create the 50 permanent DEVICE_ID records, business metadata and unique per-rig tokens; transfer tokens through the approved secret channel. **Exit:** Rig registry and credential-custody records are complete.

Step 5 — Install the edge hardware

Mount the indoor cabinet/rack equipment, connect power/UPS, HMI, managed switch, firewall and approved PLC/gateway link; update as-built records. **Exit:** Power, network, labels, earthing and physical installation checks pass.

Step 6 — Deploy the approved edge baseline

Install the finalized configuration and set only rig-specific identity, token, PLC endpoint, network values and 90-day retention. Do not redesign the tag map during rollout. **Exit:** All production containers are healthy and demo services are absent.

Step 7 — Commission rig telemetry

Enable the approved tags and spot-check critical values, units, scaling, timestamp, 1 Hz history, alarms and stale/no-data behavior against the rig PLC/HMI. **Exit:** The rig-specific commissioning sheet is signed; exceptions are disabled and recorded.

Step 8 — Connect and synchronize with CMF

Verify outbound HTTPS authentication, telemetry counts, sequence acknowledgements and data quality; perform WAN loss, local-continuity and replay checks. **Exit:** The rig is online, wrong credentials fail closed and the buffer drains after restoration.

Step 9 — Complete SAT and security acceptance

Execute the Section 7 acceptance matrix for the installed rig and associated CMF functions, including role, exposure, report, backup and resilience checks. **Exit:** No safety or critical defect remains open and all required signatories approve go-live.

Step 10 — Train, hand over and go live

Train operators and administrators; issue L1/L2/L3 runbooks, contacts, backup instructions and incident/change procedures. **Exit:** Training, operating certificate and support ownership are signed.

Step 11 — Stabilize the wave and release the next

Run wave hypercare, review health, storage, incidents, data quality, adoption and open defects, then incorporate controlled lessons into the next deployment. **Exit:** Wave KPIs are stable and governance authorizes the next wave.

5. Network, security and integration checklist

Flow	Direction	Typical port	Rule / control
PLC to collector	Edge initiates to PLC/gateway	S7 TCP 102; Modbus TCP 502; OPC UA site-defined	Read-only; allowlisted edge IP; no route from central
Edge HMI	Operator workstation to edge frontend	TCP 8080 or approved HTTPS proxy	Bind only to approved rig LAN interface; authenticated
Edge to CRMF ingest	Outbound edge to central	HTTPS TCP 443	VPN/private WAN, TLS, per-rig bearer token; gzip batches
DNS	Edge/central to approved resolver	UDP/TCP 53	Approved resolvers only
NTP	Edge/central to approved time source	UDP 123	Common authoritative time; alert on drift
AD/LDAP	Application to directory	LDAPS TCP 636 or StartTLS	Service account least privilege; certificate validation enabled
Registry / updates	Nodes to approved registry	HTTPS TCP 443	Prefer internal Harbor and signed/scanned images
Central portal	Users to ingress	HTTPS TCP 443	SSO/RBAC, WAF/rate limits, audit
Monitoring / logs	Platform to monitoring/SIEM	Site-defined	No secrets in logs; retention and alert routing approved
Database	Backend/internal only	PostgreSQL TCP 5432	Never internet/WAN exposed; network policy and least-privilege role

Cybersecurity completion checks:

- OT and IT/WAN are separate zones; only explicit flows are allowed and documented.
- No Docker socket is mounted into application containers; backend and historian are internal-only.
- Production TLS validates the issuing CA; self-signed bypass settings are prohibited outside a controlled lab.
- Real secrets are stored outside source control and rotated after any exposure; break-glass credentials are sealed and tested.
- All nodes are patched to approved baselines; images are pinned, scanned, and promoted through controlled registry stages.
- Login, configuration, alarm acknowledgement, user administration, report edits, device-token rotation, and privileged actions are auditable.

6. Configuration and data handoff checklist

Artifact	Producer	Receiver	Acceptance rule
Master tag workbook	OT / instrumentation	Edge app + QA	Addresses, types, units, scaling and ranges signed
Alarm philosophy and setpoints	Operations / process safety	Application config	Priority, delay, deadband and response approved
Network/IP/port schedule	Network architect	Rig + central teams	ACLs implemented and tested from

Artifact	Producer	Receiver	Acceptance rule
			source to target
Certificate and secret custody	Cyber / platform	Named admins	Unique, approved channel, expiry/rotation recorded
Rig master and well metadata	Operations data owner	CRMF / edge app	Stable identifiers; no duplicate rig IDs
Golden image and software bill	Platform team	Support / vendor	Hash/version recorded; restore test passed
As-built drawings and photos	Installation vendor	Asset owner	Redlines closed and approved
Test pack and defect register	QA	All stakeholders	Evidence traceable to acceptance criteria
Runbooks and escalation matrix	Support manager	L1/L2/L3 teams	On-call contacts and SLAs acknowledged

7. Minimum acceptance matrix

Area	Mandatory test	Pass criterion	Sign-off
Data acquisition	Compare agreed critical tags with PLC/HMI	Correct value, unit, scaling and timestamp; missing tags explicit	OT + operations
Live dashboard	Observe real operation and stale source	≈1 s local updates; stale/no-data shown honestly	Operations
Alarms	Trigger representative priorities	State, delay, deadband, ack and history correct	Operations + safety
Reports	Generate daily and connection records	Complete, correct date/well/rig scope; export usable	Operations
Security	Role and negative-access tests	Admin/operator/viewer boundaries enforced server-side	Cyber + QA
Offline edge	Disconnect WAN	Local HMI/historian remain operational	Rig + QA
Central sync	Outage, buffer, restore and replay	Backlog drains; central reflects full allowed interval	Network + QA
Identity	AD/LDAP/SSO and break-glass	Mapped roles correct; emergency local access controlled	IAM + app owner
Reliability	Continuous production soak	Meets agreed uptime; no uncontrolled disk growth	Support owner
Backup / DR	Restore database/configuration	Recovery meets approved RPO/RTO	Infrastructure + business continuity
Performance	Full 50-rig / <25,000 values/s production profile	Portal <5 s for normal filters; ingest and latency meet approved SLA	Performance QA
Training	Operator and admin assessment	Attendance and competency sign-off complete	Operations owner

8. Operations and support model

Level	Typical responsibility	Escalate when
L1 — Rig / NOC	Availability check, user assistance, cable/power/network checks, standard restart/runbook, incident ticket	Data stale after basic checks; repeated alarm/report issue
L2 — OT / platform	PLC/gateway diagnostics, tag/scaling, container/storage/network health, token/certificate, backup	Code defect, schema defect, security issue, unresolved data loss
L3 — Product engineering	Backend/frontend defects, database migration, performance, release and hotfix	Architecture/vendor/enterprise dependency decision
L4 — Vendor / enterprise	Hardware warranty, PLC vendor, WAN carrier, AD/PKI, storage and platform operator	Contractual or shared-service fault

- Daily: edge/central health, freshness, buffer, disk, backup job and critical security alerts.
- Weekly: data-quality score, missing tags, alarm recurrence, NPT classification, open defects and capacity trend.

- Monthly: patch/release review, restore sample, user/role review, token/certificate expiry, asset/UPS health and value metrics.
- Quarterly: DR exercise or tabletop, vulnerability review, retention/capacity forecast, architecture and rollout standard review.

9. Wave release and completion gate

A production wave may be released or closed only when all of the following are true:

- The previous wave or central foundation acceptance is signed and no safety/critical defect remains open.
- Real critical tags—not demo signals—have been validated with scaling evidence.
- Edge survives WAN outage and central replays the buffered interval correctly.
- Hardware kit, panel drawing, network design, golden image, and installation method are repeatable.
- The central platform remains within the fixed 50-rig CPU, memory, storage, ingest, backup and DR operating envelope.
- Cybersecurity approval, secret custody, audit, patching, and incident response are operational.
- Operators and support teams have signed training and handover.
- DGC/program governance has approved the wave, funding, rig list and named accountable owners.

10. Known implementation cautions from the current repository

Caution	Required action before acceptance
Repository central PVC and pod defaults are undersized for the fixed 50-rig workload	Override with Section 3 production values before deployment; monitor measured bytes/value and ingest headroom
Several central services are scaffolds or disabled	Keep out of scope or fund/build/test separately; do not count them as delivered
Historian history is not yet full min/max-envelope replay	Agree reporting/replay limitations and prioritize if operationally required
Some safety signals may differ by rig PLC implementation	Commission every critical signal at the target rig and record disabled/unavailable tags
Audit coverage at edge may not include every privileged operational action	Perform audit-gap review and close actions required by ONGC policy
Current central raw retention defaults to five years	Change to the approved 90-day raw tier plus 5-year aggregates, or fund a much larger archive tier

11. Reference basis

- Repository README.md — rig-edge architecture, local deployment, PLC integration and security model.
- central/README.md — CRMF architecture, portal modules and edge ingestion contract.
- docs/DEPLOYMENT.md — edge/central deployment, device registration, networking, signal-flow verification and operations.
- ONGC-DGC-AHWR-Digital-Twin-Proposal.md — roadmap, rig hardware, central requirements, cybersecurity and acceptance criteria.

- central/deploy/README.md and deployment manifests — RKE2 platform topology, service replicas, requests, limits, storage and DR design.
- PENDING-WORKS.md — current implementation gaps and delivery cautions.

12. Central platform terminology

This glossary explains the infrastructure terms used in the hardware and installation sections. It is intended for operational, procurement and management counterparts as well as platform engineers.

Term	Plain-language meaning	Why it matters
RKE2	A security-focused Kubernetes distribution used to run and manage CMF services across multiple servers.	Restarts failed workloads, distributes applications and supports controlled upgrades.
Kubernetes	A platform that schedules and manages containerized applications.	Provides a common operating layer across the central servers.
Control plane	The Kubernetes servers that hold cluster state and make scheduling decisions.	Three nodes provide quorum and tolerate one control-plane failure.
Worker node	A server that runs application or data workloads.	General workers run portal services; data workers run database and messaging workloads.
HA	High Availability: multiple instances and failure paths prevent a single fault from stopping service.	Required for the central 50-rig service.
PostgreSQL	The relational database engine used for operational and master data.	Stores rigs, users, alarms, governance, audit and related records.
TimescaleDB	A PostgreSQL extension optimized for timestamped telemetry.	Stores and queries rig measurements efficiently.
WAL	Write-Ahead Log: every database change is written to a recovery log before the main data files.	Enables crash recovery, standby replication and point-in-time restore.
PVC	Persistent Volume Claim: a Kubernetes request for durable storage.	Keeps database data after a container restarts or moves to another node.
CSI	Container Storage Interface: the standard link between Kubernetes and SAN, Ceph, Longhorn or another storage platform.	Allows approved enterprise storage to supply persistent volumes.
Ingress	The controlled HTTPS entry point that routes portal, API, WebSocket and ingest requests.	Gives consumers and edges one governed CMF address.
Load balancer / VIP	A redundant front door and stable virtual IP that sends traffic to healthy services.	Maintains access when one ingress or server fails.
TLS / HTTPS	Encryption and server authentication for network traffic.	Protects office and rig-to-CMF communications.
SSO	Single Sign-On using the corporate identity provider.	Lets office users use their approved enterprise account.
RBAC	Role-Based Access Control.	Limits functions and data according to viewer, operator or administrator role.
Kafka	A distributed event-streaming platform.	Optional distribution path for high-volume enterprise consumers; not required for the core portal/API.
MQTT / EMQX	MQTT is a lightweight telemetry protocol; EMQX is an MQTT broker.	Optional alternative integration route when enabled and approved.
Object storage	S3-compatible storage for backups, reports and exported files.	Keeps large objects and recovery data outside the live database.
DR	Disaster Recovery at an independent site.	Restores or assumes service if the primary data centre is unavailable.
RPO	Recovery Point Objective: maximum acceptable data loss measured in time.	Determines backup and replication frequency.
RTO	Recovery Time Objective: maximum acceptable restoration time.	Determines recovery automation and standby capacity.
Aggregate	A lower-resolution summary such as one-minute average, minimum and maximum.	Supports long retention without storing every 1 Hz value indefinitely.

13. Access for consumers at other office locations

Access principle: Office consumers access one internal CMF address through the ONGC corporate WAN or approved VPN. They must not connect to rig PLCs, rig-edge databases, the primary CMF database, Kubernetes management ports or backup administration interfaces.

Recommended traffic path: Rig edge systems send outbound HTTPS telemetry to CMF. Office users and approved applications connect to https://crmf.ongc.local over HTTPS/WSS port 443. The CMF ingress authenticates and routes requests to the portal, REST API or live WebSocket service.

Consumer	Access method	Permitted use	Mandatory control
Operations and management	CMF browser portal	Fleet overview, rig drill-down, alarms, data quality, workover, reports and governance	AD/OIDC SSO and RBAC
Live office display	Portal + secure WebSocket	Initial state from REST API; live changes through WSS	HTTPS/WSS 443 only; viewer account
Internal application	Versioned CMF REST API	Latest values, history, alarms, health, wells and reports	Dedicated read-only service identity, rate limit and audit
BI / reporting	Reporting replica, curated views or scheduled export	Power BI/Tableau and departmental reports	No queries against the primary ingest database
Data science / data lake	Controlled batch export to object storage	One-minute aggregates and approved raw intervals	Partition by rig/date; catalogue, retention and bulk-download audit
Enterprise real-time integration	Kafka topics after separate production acceptance	Multiple authorized streaming consumers	Per-consumer credentials and topic ACLs; optional work package
Platform administration	Separate privileged administration network	Kubernetes, database and recovery operations	MFA, named accounts, jump host and full audit

13.1 Office network onboarding procedure

1. **Confirm corporate connectivity:** Verify that the office is on the approved ONGC WAN/MPLS/SD-WAN or VPN and can route to the CMF application zone.
2. **Publish internal DNS:** Resolve crmf.ongc.local to the redundant CMF load-balancer VIP. Do not publish the address to the public internet.
3. **Open the minimum firewall flow:** Allow office consumer networks to the CMF VIP on HTTPS/WSS TCP 443. Do not open PostgreSQL 5432, Kubernetes, Kafka administration, MQTT administration or rig subnets.
4. **Validate certificate trust:** Ensure office devices trust the ONGC CA that issued the CMF TLS certificate.
5. **Create identity groups:** Map corporate directory groups to CMF viewer, operator and administrator roles. Use viewer as the normal office-consumer role.
6. **Onboard applications separately:** Issue a dedicated read-only service identity for each system integration; define API scope, rate limit, owner, expiry and audit requirements.
7. **Test and record access:** Confirm login, authorized screens/API functions, denied functions, session timeout and audit logging from the target office.
8.

13.2 Consumer security rules

- No office consumer is given a rig IP address or direct route to a rig edge or PLC network.
- No BI tool or ordinary application connects to the primary TimescaleDB service.
- Human users use SSO; shared user accounts are prohibited.
- System integrations use service identities with read-only scopes and documented owners.
- Bulk exports follow data-classification, retention and audit policy.
- All access is encrypted, logged and reviewed periodically.

14. Production edge installation procedure

Installation basis: Indoor edge system, fewer than 500 enabled tags, 1 Hz acquisition and 90-day local retention. Use the signed production release and a permanent unique DEVICE_ID. Never reuse a rig identity, production token or demo credential.

14.1 Prerequisites

- Installed indoor edge PC, monitor, managed switch, firewall/router and UPS meeting Section 3.
- Approved Linux LTS image, fixed IP addresses, DNS, NTP and host-hardening baseline.
- Read-only PLC/gateway route and the signed rig-specific tag workbook.
- Outbound HTTPS route to https://crmf.ongc.local and a generated per-rig device token.
- Approved offline software bundle or access to the internal signed container registry.
- Named edge administrator, operations representative and commissioning window.

14.2 Install the software

1. **Prepare the host:** Apply the approved OS patches, hostname, fixed addressing, NTP, endpoint controls, disk monitoring and host firewall. Confirm the system clock before collecting data.
2. **Install the container runtime:** Install approved Docker Engine and Docker Compose packages from the ONGC repository or offline bundle. Add only named platform administrators to the Docker administration group.
3. **Install the release files:** Create /opt/ahwr-edge, extract the signed release bundle there, verify its checksum and record the application version.

```
sudo mkdir -p /opt/ahwr-edge
cd /opt/ahwr-edge
# Extract the approved release bundle here and verify its recorded checksum.
cp .env.example .env
```

4. **Create production configuration:** Edit /opt/ahwr-edge/.env with secrets supplied through the approved secret channel. Restrict file permissions to the designated service administrator.

```
DEVICE_ID=AHWR-XX
DEVICE_TOKEN=<unique-token-issued-by-CMF>
CENTRAL_URL=https://crmf.ongc.local
SYNC_ENABLED=true
SYNC_BATCH_SECONDS=10
SYNC_BUFFER_DAYS=15
DATA_SOURCE=plc
INFLUX_RETENTION=90d
```

```
INFLUX_PASSWORD=<secret>
INFLUX_TOKEN=<secret>
JWT_SECRET=<secret>
ADMIN_PASSWORD=<secret>
FRONTEND_BIND=<approved-rig-LAN-IP>
FRONTEND_PORT=8080
AUTH_MODE=both
LDAP_URL=ldaps://<approved-directory>:636
```

5. **Validate configuration:** Review the fully resolved Compose configuration before starting. Confirm that no database or backend port is published to an unapproved interface.

```
cd /opt/ahwr-edge
docker compose config
```

6. **Start the production stack:** Use the approved production release path. The current repository builds locally; a registry-based deployment should use the approved prebuilt-image override.

```
cd /opt/ahwr-edge
docker compose up -d --build
docker compose ps
```

7. **Configure the PLC mapping:** Open the edge HMI, sign in as an administrator, configure the approved S7/Modbus/OPC UA source and apply only the signed rig-specific tags. Confirm that every connection is read-only.
8. **Configure application masters:** Configure users/roles, alarm setpoints, wells, activities, report headers, maintenance items and the approved dashboard layout.
9. **Commission local operation:** Compare live values, engineering units, timestamps and stale/no-data behavior with the PLC/HMI. Confirm alarms, reports, 1 Hz history and the 90-day retention setting.
10. **Commission CMF synchronization:** Confirm the rig is online in CMF, sequence acknowledgements increase and the edge buffer drains. Disconnect WAN, verify local continuity and buffer growth, restore WAN and confirm replay.
11. **Capture the installation record:** Save the as-built drawing, host and application versions, device identity, tag-map version, encrypted configuration backup, health evidence, SAT record and support handover.

14.3 Edge verification checklist

Check	Pass condition
Containers	All required services are running and healthy; demo-profile services are absent.
Local HMI	Authorized workstation reaches the HMI; backend and InfluxDB are not directly exposed.
Historian	1 Hz data is present and retention is 90 days.
PLC interface	Enabled tags match the signed workbook; source is read-only.
Time	PLC/gateway, edge and CMF timestamps are synchronized.
Security	Production secrets, LDAPS/TLS, roles and host firewall are active.
Offline resilience	HMI continues during WAN loss; buffered batches replay after restoration.
Backup	Configuration and persistent application state can be restored to the golden spare.

15. Production CMF installation procedure

Deployment basis: Use the HA RKE2, Helm and Ansible production path for 50 rigs. Do not use the single-host Docker Compose topology. Override the repository's development-sized pod and PVC defaults with the Section 3 production values before deployment.

15.1 Infrastructure prerequisites

- Three RKE2 control-plane nodes, three general workers and three data workers with fixed addresses and redundant 10 GbE connectivity.
- Redundant load balancer/VIP, internal DNS name crmf.ongc.local and ONGC-issued TLS certificate/issuer.
- Approved CSI storage classes providing three 5 TB database volumes, three 500 GB WAL volumes and the required Kafka volumes if enabled.
- At least 20 TB protected object/backup storage and an approved independent DR target.
- Internal Harbor registry, signed/scanned backend and frontend images, and node access to required system images.
- Vault/External-Secrets integration for database, JWT, ingest, administrator, backup and identity secrets.
- Connectivity to NTP, DNS, AD/OIDC, monitoring/SIEM, backup storage and corporate office networks.

15.2 Build the RKE2 platform

1. **Prepare deployment configuration:** Copy the example Ansible inventory and variables into site-controlled files. Populate the real control-plane VIP, node groups, hostnames, storage classes, registry, TLS issuer and non-secret environment values.

```
cd central/deploy/ansible
cp inventory.example.ini inventory.ini
cp group_vars/all.example.yml group_vars/all.yml
# Edit the site files through approved configuration control.
```

2. **Keep secrets out of source control:** Use Ansible Vault only where required during bootstrap and use Vault plus External-Secrets for runtime credentials. Do not place production passwords in committed YAML files.
3. **Apply production capacity overrides:** Before installing data services, set TimescaleDB to three instances with 5 TB data and 500 GB WAL per instance and production resource targets of 8 CPU/32 GiB requested, up to 16 CPU/64 GiB limited. Apply the approved Kafka, object-storage and DR values.
4. **Bootstrap RKE2 and operators:** Run the controlled Ansible site playbook from the administration workstation. This installs RKE2, the selected operators and the platform components defined by the repository.

```
cd central/deploy/ansible
ansible-playbook -i inventory.ini site.yml
```

5. **Confirm cluster readiness:** Verify that all control-plane and worker nodes are Ready, time is synchronized, the storage classes are available and the default-deny network baseline is active.

```
kubectl get nodes -o wide
kubectl get storageclass
kubectl get namespaces
```

15.3 Deploy data and application services

1. **Provision database secrets and backup access:** Materialize the database owner, least-privilege application login, superuser and backup credentials through External-Secrets/Vault. Ensure the backend uses the crmf_app login rather than the schema owner.
2. **Deploy TimescaleDB and platform services:** Apply the approved CloudNativePG/TimescaleDB, backup, pooler, identity, object-storage, observability and optional messaging resources. Keep scaffolded or zero-replica components disabled.
3. **Build and promote application images:** Use the approved CI pipeline to build, scan, sign and push backend/frontend images to Harbor. Pin the production release tag or immutable digest in the Helm values.
4. **Deploy the CMF application:** Install or upgrade the production Helm release with three backend replicas, three frontend replicas, HPA, PDB, NetworkPolicy, ServiceMonitor, ingress and TLS enabled.

```
helm upgrade --install crmf central/deploy/helm/crmf \
  -n crmf -f central/deploy/helm/crmf/values-prod.yaml \
  --create-namespace
```

5. **Verify Kubernetes rollout:** Confirm pods, persistent volumes, services and ingress are healthy. Verify the backend and frontend deployments complete their rollouts.

```
kubectl -n crmf get pods,pvc,svc,ingress
kubectl -n crmf rollout status deployment/crmf-backend
kubectl -n crmf rollout status deployment/crmf-frontend
```

6. **Verify internal health:** Use a controlled port-forward or approved administration path to check the backend health endpoint without publishing it externally.

```
kubectl -n crmf port-forward svc/crmf-backend 6000:6000
# From a second terminal:
curl -fsS http://127.0.0.1:6000/healthz
```

15.4 Configure CMF and onboard rigs

1. **Configure identity:** Connect AD/LDAP or OIDC, map viewer/operator/administrator groups and retain a sealed break-glass administrator.
2. **Configure production settings:** Set the 90-day raw retention tier, five-year one-minute aggregates, offline threshold, notifications, audit, dashboards, metrics and alerts.
3. **Create the rig registry:** Create all 50 stable rig identities and business metadata. Never reuse a simulator or retired DEVICE_ID.
4. **Issue per-rig credentials:** Generate one device token for each rig and transfer it through the approved secret channel. Rotate immediately if custody is uncertain.
5. **Onboard one wave at a time:** Verify telemetry counts, distinct metrics, timestamps, sequence acknowledgements, data-quality score, alarms and rig online state.
6. **Enable office consumers:** Publish internal DNS, permit HTTPS/WSS 443 from approved office networks, enable SSO groups and test portal/API access according to Section 13.

15.5 CMF production acceptance

Area	Production pass condition
HA	Loss of one control-plane, general worker, data worker or ingress instance does not stop the service.
Database	Primary and replicas are healthy; compression, retention, continuous aggregates and connection pooler are active.
Capacity	50-rig / <25,000 values-per-second profile remains inside approved CPU, memory, storage and latency thresholds.
Backup / DR	Base backup, WAL archive, restore and DR promotion meet the approved RPO/RTO.
Security	TLS, SSO/RBAC, per-rig tokens, least-privilege DB role, NetworkPolicy, secrets and audit pass.
Office access	Authorized offices reach portal/API over 443; database, Kubernetes and rig networks remain inaccessible.
Operations	Dashboards, alerts, logs, capacity thresholds, runbooks and L1/L2/L3 escalation are active.

Change control: Revise this report only through approved production change control. Record the version, approver, effective date, affected rigs/services, rollback method and change summary.